READING PASSAGE 2

*You should spend about 20 minutes on **Questions 14–26**, which are based on Reading Passage 2 below.*

Seeing in the Sea

Beneath the waves, ocean animals with bizarre and exotic ways of seeing create new understandings of vision

A Of all the human senses, the visual system – the network that turns light into neural signals that create the perception of sight – is the most studied and best understood. Our intense curiosity about sight is no coincidence: we are animals who judge the world mostly by appearances rather than, say, by aromas. But our eyes allow us only one view out of a multitude of possibilities. In the depths of the ocean – a world we are beginning to learn about in detail – marine species have developed visual systems drastically different from our own. The majority of eyes, whether on land or in the sea, share two basic components: something to focus light as it hits the eye (for humans, a cornea and lens), and something to absorb and detect the light (a retina). Then the divergences begin.

B Ocean creatures' eyes have evolved a bewildering array of structures suited to each animal's place in the natural order and the depth at which it lives. Some amphipods (deep-sea crustaceans) rely on biological fiber-optic cables to send light to their retina. Sight-dependent, nomadic predators have excellent vision, whereas many creatures that live at great depths can make only simple distinctions between light and dark. Light is abundant in the sunny shallows, but 50 feet below the surface, only the shorter wavelengths are able to penetrate, eliminating all discernable colors except blue. By about one mile down, the last traces of sunlight disappear. In this realm of perpetual night, the only illumination is manufactured, provided by the many creatures that generate their own light in a process known as bioluminescence.

C As they search for prey, swordfish swim through cold, deep water, occasionally diving as far as 3,000 feet below the surface. But even in the chilly depths, the swordfish keeps its eyes and brain 10 to 15 degrees Celsius warmer than the surrounding water by means of an unusual heat-generating muscle near the fish's eye. Vision researcher Kerstin Fritsches, of the University of Queensland in Australia, along with colleagues there and in the United States and Sweden, found that keeping warm improves the swordfish's vision. At 20°C, the scientists found, a swordfish retina can detect objects moving five times as quickly as the objects it can spot at 10°C. Since swordfish hunt speedy prey like squid, the ability to see fast-moving swimmers is crucial for finding dinner.

D The box jellyfish has surprisingly sophisticated eyes. It possesses a total of 24, divided among four eyestalks at the corners of its body. Four of the eyes on each stalk are simply groups of light-sensing cells that help it distinguish between bright and dark. The other two are complex, with lenses that produce high-quality images – but those lenses focus light well below the retina, meaning that the images are out of focus when they hit the light receptors. ‘They’re not really interested in fine details,’ says biologist Dan Nilsson, of the University of Lund in Sweden. Once the jellyfish can see the outlines of its environment, signals from its four simple brains (one in each eyestalk) steer it home to the shallow coastal waters of Australia, all the way from Puerto Rico on the other side of the world.

E The mantis shrimp, an inhabitant of tropical waters, leads a particularly vibrant life. University of Queensland neurobiologist Justin Marshall found that this crustacean has 12 different color receptors – cells in the retina that are sensitive to different wavelengths of light – whereas humans have only three. These shrimp behold a world we cannot begin to imagine: in addition to seeing a much richer light spectrum, they perceive linear and circular polarized light. Marshall suspects the ability to see polarized light ‘probably has something to do with reproduction’. A flash of polarized light reflected off a female shrimp, for example, could catch the eye of a potential mate without drawing unwanted attention from predators.

F In the deep-sea darkness, beyond the reach of the sun’s rays, some creatures make their own light. Organically created light, or bioluminescence, is so useful that it has evolved independently about 40 times across a range of creatures during the course of life on Earth, says bioluminescence expert Edith Widder, founder and CEO of the Ocean Research and Conservation Association in Fort Pierce, Florida, in the United States. Female anglerfish let someone else do the work, relying on colonies of light-emitting bacteria to give their lure its distinctive glow. The visual systems of male anglerfish are attuned to this light source; they can tell whether a female belongs to their species just by the silhouette of her lure, thus making it easy to find a mate in the dark.

G The atolla jellyfish, found in oceans across the globe, is high on beauty but low on brains. Looking beyond the scarlet tentacles will reveal that the creature has no central nervous system. Since it lacks the neural architecture to interpret complex visual information, the jellyfish must rely on its simple but numerous pigment cups. When attacked, as part of its defence, the atolla launches a stunning blue bioluminescent display visible for hundreds of feet through the dusky mid-depth waters where it lives. Blue hues are the only colors of light able to penetrate that far below the surface, so the atolla’s red pigmentation normally makes it largely invisible.

Questions 14–19

Reading Passage 2 has seven paragraphs.

Which paragraph contains the following information?

*Write the correct letter, **A–G**, in boxes 14–19 on your answer sheet.*

***NB** You may use any letter more than once.*

14 an example of an ocean animal which can see colour differently from humans

15 a simple description of how most eyes work

16 a reference to the frequency with which one biological trait has developed in different species

17 a description of a sea creature which travels huge distances

18 details of a creature that protects itself by means of light

19 a description of a creature with multiple eyes

Questions 20 and 21

*Choose **TWO** letters, **A–E**.*

Write the correct letters in boxes 20 and 21 on your answer sheet.

In which **TWO** of the following sea creatures does one gender attract the other using light?

A swordfish
B box jellyfish
C mantis shrimp
D anglerfish
E atolla jellyfish

Questions 22–26

Complete the summary below.

*Choose **ONE WORD AND/OR A NUMBER** from the passage for each answer.*

Write your answers in boxes 22–26 on your answer sheet.

How the eyes of ocean creatures have developed differently from those of land creatures

Ocean creatures' eyes have developed many different **22** ..... according to the depth of the water where they live. Ocean roaming predators have well developed eyesight because they hunt where there is plenty of light. However, light only reaches the **23** ..... of some sea creatures of the deep via fibre-optic cables, and by 50 feet deep, **24** ..... is the only colour which is visible. At a depth of around **25** ....., no light from the sun can get through. At this level, the only source of light is that which is produced through **26** .....

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Seeing in the Sea

01 段落信息匹配

海中视觉 · 逐题精析

题目要求：从 A-G 段中找出对应信息所在段落；同一字母可以使用多次。

14 E　15 A　16 F　17 D　18 G　19 D

题号	题干翻译与答案	完整定位句与翻译	详细解析与易错点
14	**答案 E** an example of an ocean animal which can see colour differently from humans 举出一种海洋动物的例子：它感知颜色的方式与人类不同。	**E段第2句｜螳螂虾与人类的对比** University of Queensland neurobiologist Justin Marshall found that this crustacean has 12 different color receptors - cells in the retina that are sensitive to different wavelengths of light - whereas humans have only three. 译：昆士兰大学神经生物学家贾斯廷·马歇尔发现，这种甲壳动物有12种不同的颜色感受器，即视网膜中对不同光波波长敏感的细胞；而人类只有3种。	**比较对象必须同时出现** this crustacean 指本段开头的 mantis shrimp（螳螂虾）。句中将它的12种颜色感受器与人类的3种直接比较；whereas 引出差异，落实题干的 differently from humans。 颜色感受器能够感知不同波长的光，种类的差异说明两者的颜色视觉不同。B段主要说明水深如何限制可见颜色，没有把某种动物与人类直接比较。
15	**答案 A** a simple description of how most eyes work 简要描述大多数眼睛的工作方式。	**A段倒数第2句｜眼睛共有的两项功能** The majority of eyes, whether on land or in the sea, share two basic components: something to focus light as it hits the eye (for humans, a cornea and lens), and something to absorb and detect the light (a retina). 译：无论在陆地上还是海里，大多数眼睛都有两个基本组成部分：一个在光进入眼睛时将其聚焦（对人类而言，是角膜和晶状体），另一个吸收并感知光（即视网膜）。	**由结构读出工作过程** the majority of eyes 对应 most eyes。冒号后的两项依次说明：先 focus light（聚焦光线），再 absorb and detect the light（吸收并感知光线）。这就是对眼睛基本工作方式的概括。 题干限定的是“大多数眼睛”。D段细讲箱水母，涉及单一动物，范围不符合 most。
16	**答案 F** a reference to the frequency with which one biological trait has developed in different species 提到某一生物特征在不同物种中演化出现的次数。	**F段第2句｜生物发光的独立演化** Organically created light, or bioluminescence, is so useful that it has evolved independently about 40 times across a range of creatures during the course of life on Earth, says bioluminescence expert Edith Widder, founder and CEO of the Ocean Research and Conservation Association in Fort Pierce, Florida, in the United States. 译：生物体产生的光，即生物发光，非常有用，因此在地球生命发展的过程中，它已在多种生物中独立演化了约40次。美国佛罗里达州皮尔斯堡海洋研究与保护协会的创始人兼首席执行官、生物发光专家伊迪丝·威德这样说。	**frequency 在这里指演化次数** one biological trait 指 bioluminescence（生物发光）；has developed 对应 has evolved；different species 对应 a range of creatures。about 40 times 给出所问的次数。 independently 表示该特征曾分别、独立地演化出现。“40次”不能改读成“40个物种”，也没有说明某只动物一天发光多少次。
17	**答案 D** a description of a sea creature which travels huge distances 描述一种会跨越极远距离的海洋生物。	**D段末句｜箱水母的长距离移动** Once the jellyfish can see the outlines of its environment, signals from its four simple brains (one in each eyestalk) steer it home to the shallow coastal waters of Australia, all the way from Puerto Rico on the other side of the world. 译：一旦水母能看清周围环境的轮廓，来自它四个简单脑结构（每个眼柄中有一个）的信号，就会引导它从世界另一端的波多黎各一路返回澳大利亚沿岸的浅水水域。	**起点和终点共同说明距离** all the way from Puerto Rico 表示“从波多黎各一路而来”；目的地是 Australia。on the other side of the world 强调二者相距极远，对应 huge distances。 G段 across the globe 说的是分布范围广，不能推出某一个体移动了很远。C段的3,000英尺是下潜深度；D段的跨地域行程更直接对应题干。
18	**答案 G** details of a creature that protects itself by means of light 说明一种借助光来保护自己的生物。	**G段倒数第2句｜阿托拉水母的防御** When attacked, as part of its defence, the atolla launches a stunning blue bioluminescent display visible for hundreds of feet through the dusky mid-depth waters where it lives. 译：受到攻击时，作为防御的一部分，阿托拉水母会发出耀眼的蓝色生物光。在它生活的昏暗中层水域，这种光在数百英尺之外仍可被看见。	**核对发光的用途** When attacked 交代受到攻击的情境；as part of its defence 明确说明防御目的，对应 protects itself。blue bioluminescent display 是使用的手段，对应 by means of light。 E、F段也提到光，但相关信息指向繁殖和寻找配偶。不能看到 light 就选，必须同时核对“保护自己”这一目的。
19	**答案 D** a description of a creature with multiple eyes 描述一种拥有多只眼睛的生物。	**D段第1-2句｜箱水母的眼睛数量** The box jellyfish has surprisingly sophisticated eyes. It possesses a total of 24, divided among four eyestalks at the corners of its body. 译：箱水母拥有复杂程度令人惊讶的眼睛。它共有24只眼睛，分布在身体各角的四个眼柄上。	**省略的名词要从前句补出** a total of 24 后省略了 eyes；前一句已经提供该名词。因此这里明确说有24只眼睛，直接对应 multiple eyes。 four eyestalks 指四个眼柄，眼柄是承载眼睛的结构，不能将眼柄数当作眼睛数。E段的12种 color receptors 是感受器种类，也不等于12只眼睛。D段可与第17题重复使用。

段落定位速览

A 共同视觉结构　B 水深与光线　C 剑鱼保温与捕猎　D 箱水母的眼睛与导航　E 螳螂虾的色觉与求偶　F 生物发光与鲛鳚鱼求偶　G 阿托拉水母的防御

Seeing in the Sea

02 双选题

海中视觉 · 逐题精析

题目要求：从 A-E 中选出两项，分别填入第20、21题；两个字母顺序可互换。

20 C　21 D　｜　C、D 顺序可互换

In which TWO of the following sea creatures does one gender attract the other using light?

以下哪两种海洋生物会由一个性别借助光吸引另一个性别?

选项	名称与判断	完整定位句与翻译	入选或排除的具体理由
A	**swordfish** 剑鱼 **排除**	**C段末句｜视力服务于捕猎** Since swordfish hunt speedy prey like squid, the ability to see fast-moving swimmers is crucial for finding dinner. 译：由于剑鱼会捕食鱿鱼这类游动迅速的猎物，因此，看清快速游动的动物的能力，对它找到食物至关重要。	本段描述眼部附近的肌肉产生热量，使眼睛和脑保持较高温度，从而提高追踪快速猎物的能力。末句明确说明用途是 finding dinner（找到食物）。 heat-generating 是“产热”，不能读成“发光”。文中未将剑鱼的这种能力与异性吸引联系起来。
B	**box jellyfish** 箱水母 **排除**	**D段末句｜感知轮廓以导航** Once the jellyfish can see the outlines of its environment, signals from its four simple brains (one in each eyestalk) steer it home to the shallow coastal waters of Australia, all the way from Puerto Rico on the other side of the world. 译：一旦水母能看清周围环境的轮廓，来自它四个简单脑结构（每个眼柄中有一个）的信号，就会引导它从世界另一端的波多黎各一路返回澳大利亚沿岸的浅水水域。	本句中 see the outlines 的作用是识别环境轮廓，随后通过 signals 引导移动。steer it home 明确指向导航、返回栖息水域。 “能够感知光和环境”不自动包含“用光吸引异性”。本段没有交代雌雄之间借光寻找配偶的行为。
C	**mantis shrimp** 螳螂虾 **入选｜第20题**	**E段末两句｜偏振光与潜在配偶** Marshall suspects the ability to see polarized light 'probably has something to do with reproduction'. 译：马歇尔推测，感知偏振光的能力“可能与繁殖有关”。 A flash of polarized light reflected off a female shrimp, for example, could catch the eye of a potential mate without drawing unwanted attention from predators. 译：例如，雌性螳螂虾身上反射出的一束偏振光，可能会吸引潜在配偶的注意，同时不会招来捕食者的不必要关注。	female shrimp（雌虾）身上反射偏振光，potential mate（潜在配偶）注意到它，两种性别之间通过光发生吸引，满足题干条件。 catch the eye of 对应 attract，reproduction 提示繁殖语境。using light 包括利用反射光，题干没有限定必须自己产生光。 原文用 suspects、probably、could 保留推测语气，解析也应保留；不要把它扩大成“雌虾每次都会发光求偶”。
D	**anglerfish** 鲛鳚鱼 **入选｜第21题**	**F段末两句｜发光细菌与雌雄识别** Female anglerfish let someone else do the work, relying on colonies of light-emitting bacteria to give their lure its distinctive glow. 译：雌性鲛鳚鱼让其他生物代劳，依靠成群的发光细菌使自己的诱饵状器官发出独特的光。 The visual systems of male anglerfish are attuned to this light source; they can tell whether a female belongs to their species just by the silhouette of her lure, thus making it easy to find a mate in the dark. 译：雄性鲛鳚鱼的视觉系统对这种光源十分敏感；只凭雌鱼诱饵状器官的轮廓，它们就能判断雌鱼是否与自己属于同一物种，从而容易在黑暗中找到配偶。	雌鱼的诱饵状器官发光，雄鱼对这一光源敏感，并据其轮廓识别同种雌鱼；find a mate 直接说明寻找配偶的用途。 female 与 male 对应 one gender 与 the other；this light source 指上句诱饵状器官发出的光。光由细菌产生，仍属于鱼借助光完成异性识别。 someone else 在下文落实为 bacteria（细菌），不能理解成“另一条鱼”。
E	**atolla jellyfish** 阿托拉水母 **排除**	**G段倒数第2句｜受攻击时发光** When attacked, as part of its defence, the atolla launches a stunning blue bioluminescent display visible for hundreds of feet through the dusky mid-depth waters where it lives. 译：受到攻击时，作为防御的一部分，阿托拉水母会发出耀眼的蓝色生物光。在它生活的昏暗中层水域，这种光在数百英尺之外仍可被看见。	原文明示发光的用途是 defence（防御），触发情境是 attacked（受到攻击）。这可支持第18题，却不满足本题的异性吸引条件。 “确实会发光”和“发光用于求偶”是两层信息。选择时必须找到后一层，不能仅凭 bioluminescent 入选。

双选题核对方法

把三个条件放在同一条关系里核对：雌雄双方＋光的参与＋吸引或寻找配偶。C项由 female、potential mate、catch the eye of 建立关系；D项由 female、male、find a mate 建立关系。

Seeing in the Sea

QUESTIONS 22-26

03 摘要填空

海中视觉 · 逐题精析

题目要求：每空从原文选取 ONE WORD AND/OR A NUMBER（一个单词和／或一个数字）。

22 structures　23 retina　24 blue　25 one mile / 1 mile　26 bioluminescence

How the eyes of ocean creatures have developed differently from those of land creatures

摘要标题：海洋生物的眼睛在演化上与陆地生物有何不同。

题号	含空题干、翻译与答案	完整定位句与翻译	同义替换、句法与易错点
22	**答案 structures** Ocean creatures' eyes have developed many different 22 _____ according to the depth of the water where they live. 海洋生物的眼睛根据其生活水域的深度，演化出了许多不同的_____。	**B段第1句｜结构适应生存环境** Ocean creatures' eyes have evolved a bewildering array of structures suited to each animal's place in the natural order and the depth at which it lives. 译：海洋生物的眼睛已演化出种类繁多、令人眼花缭乱的结构，这些结构适应各自动物在自然界中的位置及其生活的深度。	developed 对应 evolved；many different 对应 a bewildering array of；according to the depth 对应 suited to... the depth。三组信息共同指向 structures（结构）。 many different 后面需要可数名词复数，所以保留复数词尾s。空格问“演化出了什么”，depth 已在后半句充当影响因素，不能填入。
23	**答案 retina** However, light only reaches the 23 _____ of some sea creatures of the deep via fibre-optic cables 然而，对于一些深海生物，光只能通过光纤到达它们的_____。	**B段第2句｜光经光纤到达视网膜** Some amphipods (deep-sea crustaceans) rely on biological fiber-optic cables to send light to their retina. 译：某些端足类动物（深海甲壳动物）依赖生物光纤，将光输送到它们的视网膜。	send light to their retina 改写成 light reaches the retina，句子的主语由动物变为 light，终点仍然是 retina（视网膜）。via 表示“通过”，对应 rely on... cables 表达的传输途径。 the 后接表示身体部位的名词，直接沿用原文单数 retina。cables 已写在题干里，表示路径；空格问的是光到达的部位。
24	**答案 blue** and by 50 feet deep, 24 _____ is the only colour which is visible. 而到了50英尺深处，_____是唯一可见的颜色。	**B段第4句｜蓝色之外的颜色消失** Light is abundant in the sunny shallows, but 50 feet below the surface, only the shorter wavelengths are able to penetrate, eliminating all discernable colors except blue. 译：阳光照耀的浅水区光线充足，但在水面以下50英尺处，只有较短波长的光能够穿透，除蓝色外，其他所有可辨认的颜色都消失了。	all... except blue 表示“除蓝色之外的全部颜色”。这些颜色都被排除后，剩下的可见颜色就是 blue，对应 the only colour。 blue 在本句中作表示颜色的名词，填入后形成 blue is the only colour。shorter wavelengths 解释光的波长特征，不是所问的颜色名称。
25	**答案 one mile / 1 mile** At a depth of around 25 _____, no light from the sun can get through. 在大约_____的深度，太阳光完全无法穿透。	**B段第5句｜最后一丝阳光消失** By about one mile down, the last traces of sunlight disappear. 译：下潜到大约一英里深时，最后残留的阳光也消失了。	around 对应 about；no light from the sun can get through 对应 the last traces of sunlight disappear。答案取自 one mile，可写 one mile 或 1 mile。 one 与 1 都表示数量“一”，mile 是单位；答案须保留“数量＋单位”，mile 保持单数。空格后没有给出单位，仅写 one 或 1 无法完整表达深度。 50 feet 对应上一题“仅蓝色可见”的深度；本题要求“阳光完全消失”，须定位下一句。
26	**答案 bioluminescence** At this level, the only source of light is that which is produced through 26 _____. 在这一深度，唯一的光源是通过_____产生的光。	**B段末句｜黑暗中的生物发光** In this realm of perpetual night, the only illumination is manufactured, provided by the many creatures that generate their own light in a process known as bioluminescence. 译：在这个永远黑暗的区域，唯一的照明是生物产生的，由许多能够自行发光的生物提供；它们产生光的过程称为生物发光。	At this level 承接上句约一英里的深度；that which is produced 指所产生的光。through 后需要一个表示产生方式或过程的名词，原文的 process known as 直接引出 bioluminescence（生物发光）。 manufactured 在此结合后文指生物产生光，不能译为人类制造的“人工照明”。bacteria 仅是F段雌性鲛鳐鱼的具体光源，本空要求B段给出的过程名称。

摘要中的衔接句

Ocean roaming predators have well developed eyesight because they hunt where there is plenty of light.

译：在海洋中四处游动的捕食者拥有发达的视力，因为它们在光线充足的地方捕猎。

本组五个答案全部位于B段，顺序为：眼睛的结构、接收光的部位、50英尺处可见的颜色、阳光消失的深度、深处的发光过程。第23、24题分处同一句的两个并列分句。